Li-Kang (Tony) Weng

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WORK EXPERIENCE

Software Development Engineer, Amazon Web Services, Inc. *Seattle, USA | 06/2021 – present*

- Own tier-1 EC2 Fleet launch APIs serving thousands of TPS at **five-nine** availability across all AWS regions.
- Tech lead for the fleet-provisioning component of EC2's demand-shaping program; drove design alignment and delivery across **20+** partner teams, contributing **billion-dollar-scale** incremental CapEx savings.
- Led the cross-team design and delivery of the public EC2 Capacity Blocks instant-launch feature on a fixed re:Invent deadline, cutting customer **time-to-GPU-capacity from hours to immediate**.
- Built an internal AI builder that grounds code generation in team-specific context indexed from internal engineering documentation, producing implementations that match team conventions instead of generic output requiring rework.
- Replaced heavy manual operational workflows — daily ticket triage, status aggregation, and audit reporting — with LLM-based automation, cutting **$3M+ OpEx saving** and reclaiming **1+ engineer-month** per year.
- Mentored 10+ engineers across organizations; several promoted.

Software Development Engineer Intern, Amazon Web Services, Inc. *Dallas, USA | 05/2020 – 07/2020*

- Added CloudWatch Events support to EC2/Spot Fleet so customers could observe fleet state changes and react programmatically.

Research Assistant, National Taiwan University *Taipei, Taiwan | 03/2019 – 06/2019*

- Built a tracked surveillance robot on ROS for a poultry farm, with telemetry and remote control surfaced through a custom GUI.

Software Engineer, HTC Corp. *Taipei, Taiwan | 01/2017 – 07/2018*

- Developed a real-time visual-inertial SLAM algorithm in C++ on an ARM platform.
- Built visualization and KPI measurement tooling that replaced manual inspection and made tracking performance quantitatively comparable across builds.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, C++, JavaScript, Ruby

Infrastructure: AWS (EC2, DDB, Lambda, S3, CloudWatch), high-availability distributed systems

AI: LLM-based automation, retrieval over internal documentation, context-grounded code generation, prompt engineering

EDUCATION

The University of Texas at Dallas — M.S. Computer Science *Dallas, USA | 05/2021*

National Taiwan University — M.S. and B.S., Bio-Industrial Mechatronics Engineering *Taiwan | 09/2015, 06/2013*

EARLIER RESEARCH AND AWARDS

- **Sensor fusion for vehicle safety (NTU BBLab / ARTC, 2012–2015):** built a real-time obstacle detection, tracking, and collision-avoidance system fusing stereo vision with millimeter-wave radar. Reduced depth measurement error from 2.4% to 0.7%, raised obstacle match rate from 82.1% to 89.8%, and accelerated correspondence matching 2.8x using CUDA and OpenCV.
- **2nd prize, 9th Utechzone Machine Vision Prize (2014):** fall-detection algorithm robust to lighting variation and occlusion, combining background removal, feature extraction, object tracking, and motion detection.
- **Publication:** Ta-Te Lin, Li-Kang Weng, An-Chih Tsai. "Object Tracking and Collision Avoidance Using Particle Filter and Vector Field Histogram Methods." ASABE 2014, Paper No. 1906189, Montreal, Canada.